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Pratheek Prakash Shetty

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GPU systems and compiler researcher focused on ML compiler infrastructure and performance optimization for parallel and heterogeneous architectures.

SKILLS

- Proficient in: Python, C++, C, Rust, SQL, TypeScript
- Parallel Computing: CUDA, ROCm, OpenMP, SYCL, Multithreading, HPC, Nsight Compute, rocprof
- Systems: UNIX/Linux, Operating Systems, Computer Architecture, concurrent data structures, lock-free/wait-free algorithms
- Compilers/Runtimes: LLVM IR, MLIR, Triton, PyTorch, compiler optimization, data-flow analysis, tiling, fusion, autotuning

EDUCATION

Master of Science, Computer Engineering Aug 2024 - May 2026 (Expected)
Virginia Polytechnic Institute and State University

- Coursework: Compiler Optimization, Computer Vision, Computer Architecture, Multi-processor programming, Advanced Parallel Computation.

Bachelor of Engineering, Computer Science and Engineering Aug 2019 - Aug 2023
Acharya Institute of Technology, Bengaluru

RESEARCH EXPERIENCE

Research Assistant, Synergy Lab, Virginia Tech Dec 2024 - Present

- **Wait-Free Concurrent Queue Design** - Implemented and designed novel GPU wait-free queues in C++; profiled and evaluated on BFS and Ray Tracing application, achieving 2-5x speedup over baselines.
- **GPU Atomics Characterization** - Analyzed atomic operation performance across modern NVIDIA, AMD and Intel GPUs using architecture specific profiling tools (Nsight Compute, rocprof, VTune); studied contention patterns, memory coherence, and concurrent data structure behavior across different GPU architectures.
- **CoreTSAR-M SYCL/oneAPI Runtime**: Researching a task-distribution runtime for heterogeneous workloads; designed a persistent device-side scheduling algorithm in C++/SYCL that uses wait-free concurrent queues to keep scheduling device resident.

PROFESSIONAL EXPERIENCE

Software Engineer, Virginia Tech Sep 2024 - May 2025

- Building a real-time data acquisition and visualization system to enable researchers to capture high-frequency signals (10MHz+) using Electron, React and Python.
- Implemented multi-threaded data processing pipeline, increasing throughput by 2.5x.

SDE Intern, Instahyre Mar 2024 - Jun 2024

- Restructured microservices architecture for 3x more concurrent users, reducing API response time by 40% while optimizing Django Backend decreasing average query time from 1.2s to 0.3s.

PROJECTS

Tvastr: Contract-Driven GPU DSL Prototype Jun 2026 - Present

- Building a prototype GPU DSL generator that converts tensor-operation contracts into PyTorch references, Triton kernel scaffolds, runtime guards, schedule spaces, autotuning scripts, and MLIR/TableGen stubs.
- Implemented shape/dtype validation, layout guards, rewrite legality checks, schedule pruning, and heuristic schedule ranking for a fused `matmul + bias + ReLU` kernel.

Autotiling Pass for LLVM/MLIR March 2025 - May 2025

- Built a C++ MLIR pass that analyzes data reuse patterns, selects cache-optimal tile sizes, fuses loops, and inserts prefetches for GPU kernels; achieved $1.9\times$ speedup on GEMM/Conv with 2% compile overhead

RISCV Sine Hardware Accelerator Nov 2024 - Dec 2024

- Designed a custom RISC-V hardware accelerator for trigonometric workloads in graphics and physics-style kernels; reduced cycle count by 43% compared with a software implementation while maintaining precision.

Rlu-BplusTree: BPlusTree in rust for performance and safety Sep 2024 - Dec 2024

- Created concurrent B+ tree implementation to address data race issues in multi-threaded database systems.
- Achieved 90% of C implementation speed while eliminating memory safety vulnerabilities present in C version.